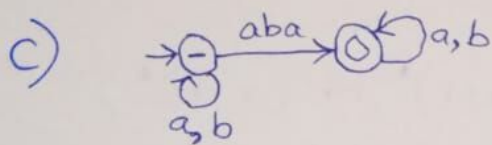
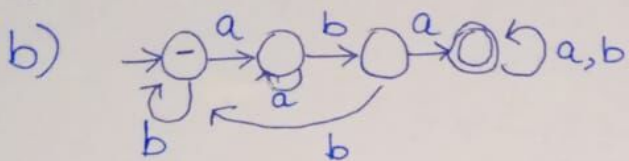
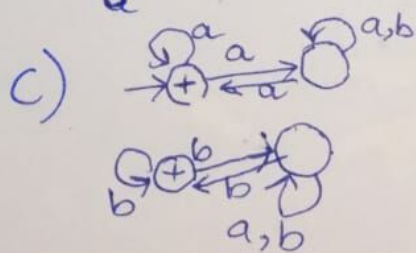
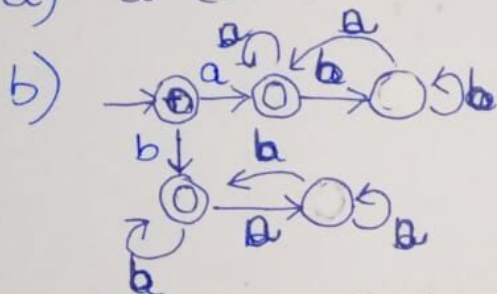


QUESTION 1

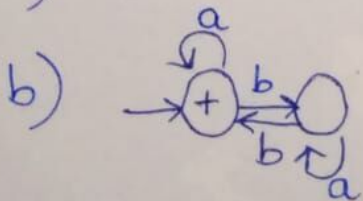
i) a) $(a+b)^* aba (a+b)^*$



ii) a) $a (a+b)^* a + b (a+b)^* b + a + b + \lambda$

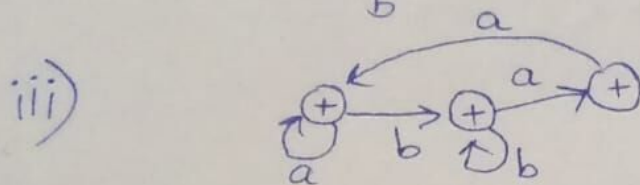
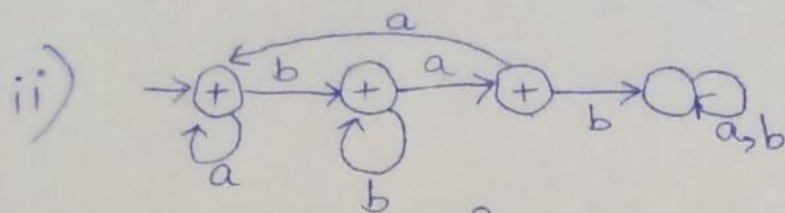


iii) a) $(a^* b a^* b a^*)^*$

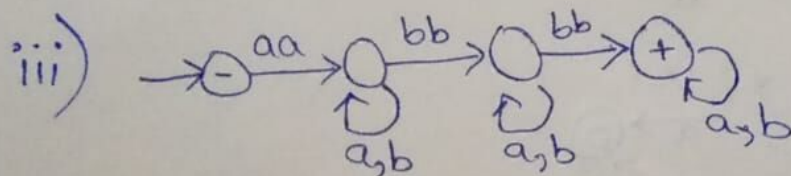
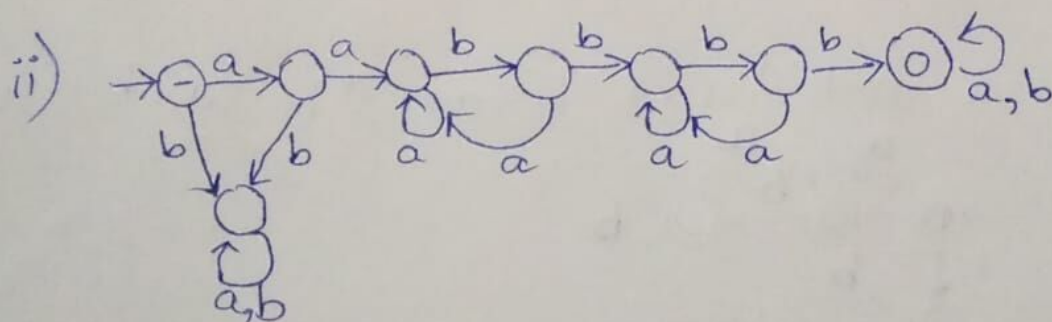


c) FA = TG

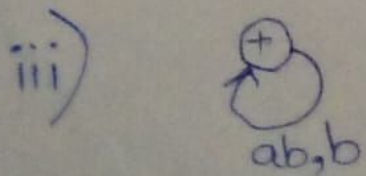
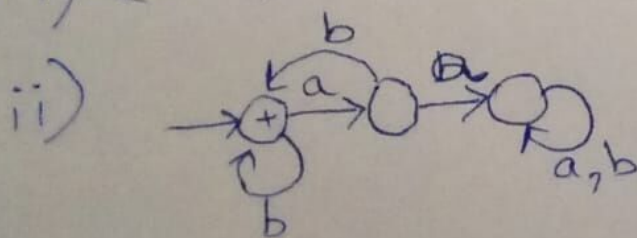
iv) i) $(a^*bb^*aa)^*a^*(b+bb^*+bb^*a+\lambda)$



v) i) $aa((a+b)^*bb(a+b)^*bb(a+b)^*)$

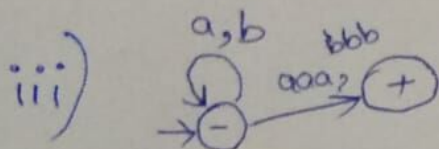
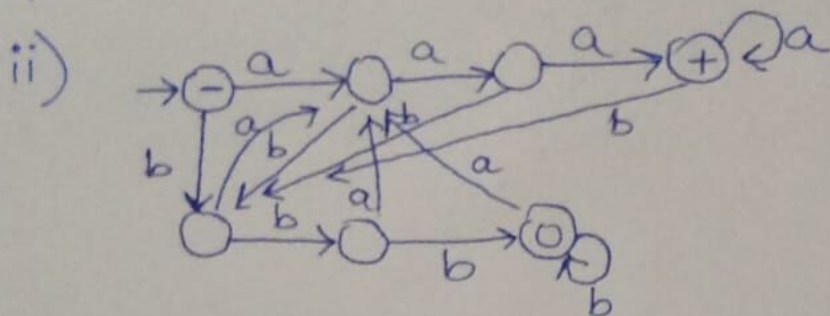


vi) i) $(ab+b)^*$

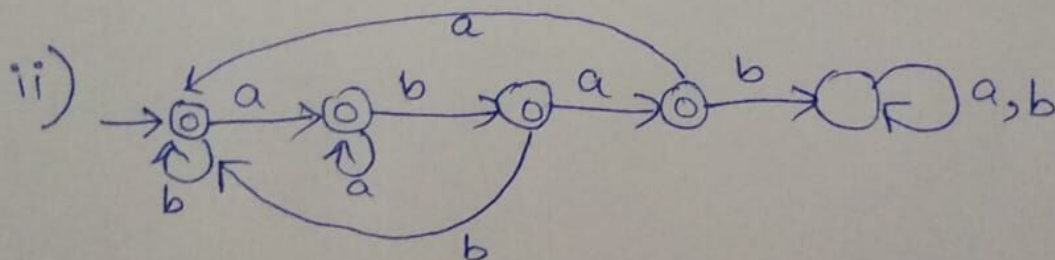


vii) Cannot be expressed in terms of RE.
 → RE is not used in counting
 → Difficult to keep track of numbers.

viii) i) $(a+b)^* (aaa + bbb)$



ix) i) $((b^* a a^* (b b + b a a)^*)^*)^* b^* (a + a a^* + a a^* b + a a^* b a + \lambda)$



iii) Everything's same except for deadstate we can drop it in TG.

x)



iii)

QUESTION 2

- i) Strings containing at least one "aa".
- ii) Even length strings that do not contain more than one "aa" or "bb" together.
- iii) Language containing b's within a's or either sides or no b's at all.
- iv) Strings containing Even number of b's.
- v) Strings containing at least one a followed by pattern $b^*ab^*a^*$ containing a's or bab's or b'sa b's a's.

QUESTION 3

I. Equivalency of $((a+bb)^*aa)^*$ & $(a+bb)^*aa+\lambda$

i) $((a+bb)^*aa)^*$

- contains λ
- strings end with aa
- any number of a's and bb's
- even number of b's
- always double b's.

ii) $(a+bb)^*aa+\lambda$

- contains λ
- strings end with aa
- any number of a's and bb's

- even number of b's
- always double b's

in (I) whole Kleen closure was necessary for λ only since all combos can be made with the inner Kleen closure as well.

in (II) λ is introduced separately,

Therefore both are equivalent!

II. Equivalency of $a(ba+a)^*b$ & $aa^*b(aa^*b)^*$

i) $a(ba+a)^*b$

- starts with a, ends with b
- any number of a's & ba's in between
- no consecutive b's

ii) $aa^*b(aa^*b)^*$

- starts a, ends b
- any num of (ba)'s and a's in between
- no two b's are consecutive

Considering above outcomes, the REs define the same Language and are equivalent

QUESTION 4

I. Recursive Definition:-

$$L = \{ \lambda, ab, aa, bb, ba, aaaa \dots \}$$

- i) Base Step: λ is in the language L
- ii) Recursive Step: if x is in L , then so are:

$$\begin{array}{l} axa, axb \\ bxa, bxb \end{array}$$

- iii) Exceptional case: No strings except for those constructed in above, are allowed to be in the L .

II. Recursive Definition:-

$$L = \{ a, b, aba, bab, aaa \dots \}$$

- i) Base Step: a, b is in the L .
- ii) Recursive Step: if x is in the L , then so are,

$$axa, bxb$$

- iii) Exceptional cases:

No strings except for those constructed in above, are allowed to be in the L .

III. Recursive Definition :-

$$L = \{a, b, aaa, baa, aba, \dots\}$$

- i) Base Step: a, b are in L .
- ii) Recursive Step: if x is in L ,
then so ax ,
 bxb , axa , axb , bxa
- iii) Exceptional cases:
No strings except for those constructed
in above, are allowed to in the
Language L .